

The screenshot shows the Jenkins web interface. The left sidebar contains links: Back to Project, Status, Changes, Console Output (selected), View as plain text, Edit Build Information, Delete build '#9', Git Build Data, No Tags, and Previous Build. The main area is titled 'Console Output' and shows the following text:

```

Started by user Manav Desai
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/start-model-server
The recommended git tool is: NONE
using credential c9767011-0a6e-4bcd-ab25-d286c5afa218
> git rev-parse --is-inside-work-tree # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/mkd1997/taxi-pipeline-client # timeout=10
Fetching upstream changes from https://github.com/mkd1997/taxi-pipeline-client
> git --version # timeout=10
> git --version # 'git version 2.25.1'
using GIT_ASKPASS to set credentials
> git fetch --tags --force --progress -- https://github.com/mkd1997/taxi-pipeline-client
+refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision 23b0f82786b9e1f5229aef4fd3e0ac7ae53b77f5 (refs/remotes/origin/master)
> git config core.sparsecheckout # timeout=10
> git checkout -f 23b0f82786b9e1f5229aef4fd3e0ac7ae53b77f5 # timeout=10
Commit message: "modify script"
> git rev-list --no-walk ba2140e3678dd3b65adedc4abc0578500c5da745 # timeout=10
[start-model-server] $ /bin/sh -xe /tmp/jenkins3985344148914038186.sh
+ pwd
+ bash start_model_server_local.sh /var/lib/jenkins/workspace/start-model-server/resources/model:
Pulling the Docker image: tensorflow/serving

```

Figure 6: Output of running the Jenkins job

sends them to the server container. It receives a list of class predictions in JSON format, as shown in Figure 7.

Model deployment to the server can be done manually as well as by using Jenkins. We have explored both options in this article. To run the script manually, enter the following command in a terminal:

```
$bash <path to start_model_server_local.sh> <path_to_model_directory>
```

The output is shown in Figure 5.

The model deployment using the bash script can also be done using Jenkins. To do so, first initialise a Git repository in the folder containing the server script, and then push the repository to GitHub. Also add the model and schema files to the repository. Then, start the Jenkins service and open the Jenkins dashboard in a browser. Create a new Freestyle project and enter the configuration fields as follows:

- **Source code management:** Git
- **Repositories:** URL to your GitHub repository

Then add a build step to execute shell and in the command section, enter the following command:

```
$bash start_model_server_local.sh
<path_to_model_directory>
```

The console output of the Jenkins job is shown in Figure 6.

Now, to run inference on the deployed model and to run the client script, give the following command:

```
$bash <path to classify_local.sh> <path
to data file> <path to schema file>
```

The output of running the inference on three examples is shown in Figure 7.

Since the problem that this project deals with comprises binary classification – predicting whether the passenger will tip the driver or not, given the details about the taxi ride, we get class probabilities as output for each example sent as a request. The result sent is in the form of a JSON object, which in turn consists of a list of JSON objects. Each individual JSON object

```

{
  "predictions": [
    {
      "scores": [
        0.589885,
        0.410115063
      ],
      "classes": [
        "0",
        "1"
      ]
    },
    {
      "scores": [
        0.758405805,
        0.24159427
      ],
      "classes": [
        "0",
        "1"
      ]
    },
    {
      "scores": [
        0.859367,
        0.140632957
      ],
      "classes": [
        "0",
        "1"
      ]
    }
  ]
}

```

Figure 7: Running inference on the model

has two parameters – list of output classes and also a list of individual class probabilities. The final output